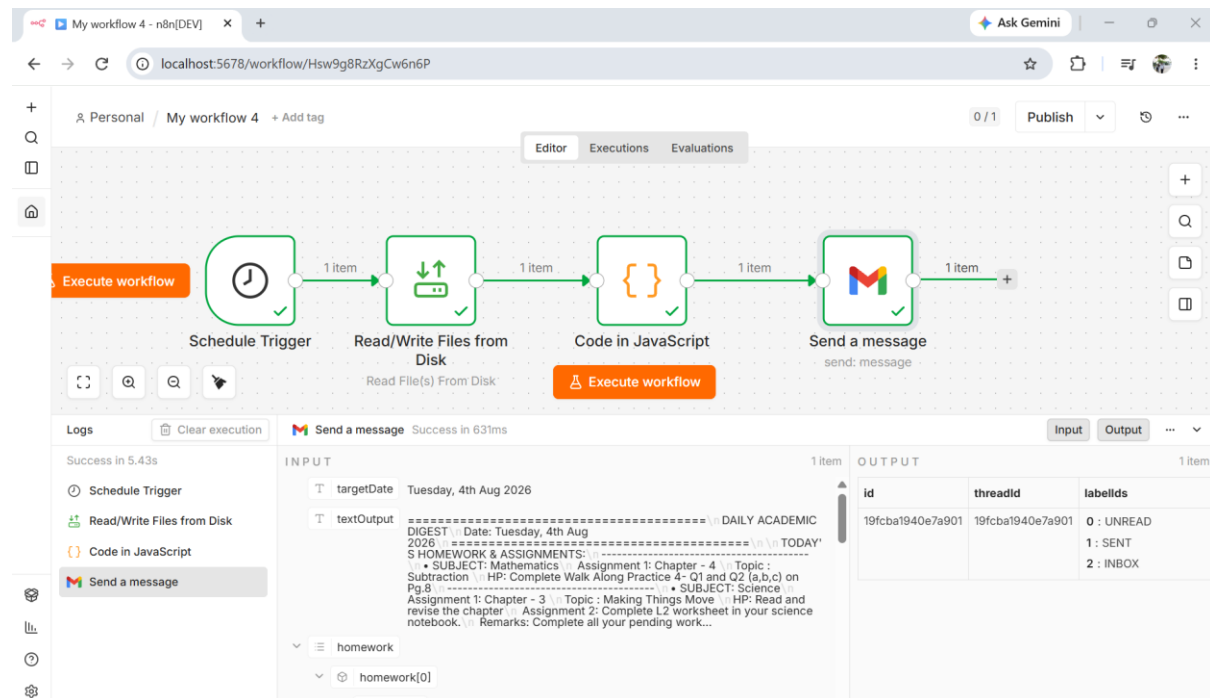


Automated Daily Academic Schedule & Homework Dispatcher

1. Objective

The primary goal of this workflow is to eliminate the manual effort required to track daily school assignments, schedules, and requirements. Instead of manually opening and searching through a comprehensive master Excel spreadsheet every day, this automation extracts only the relevant information for the current date and delivers a clean, beautifully formatted HTML digest directly to the designated recipients' email inboxes.

2. Workflow Architecture (Implementation)



This automation is built using **n8n**, a node-based workflow automation tool. The pipeline consists of four sequential nodes that handle scheduling, file retrieval, data processing, and communication.

- **Node 1: Schedule Trigger**
 - **Function:** Acts as the chronometer for the workflow.
 - **Configuration:** Set to execute automatically at a specific time every day (e.g., 5:00 PM), ensuring the digest is ready for evening homework sessions.
- **Node 2: Read/Write Files from Disk**
 - **Function:** Interacts with the local file system (or network drive) to retrieve the raw data.
 - **Configuration:** Reads the master academic .xlsx file and outputs it as a binary data buffer for the next node to process.
- **Node 3: Code (JavaScript)**
 - **Function:** The core processing engine of the workflow.

- **Configuration:** Uses the `xlsx` library to parse the binary Excel data. It dynamically locates today's date within the sheet, extracts relevant subjects, assignments, requirements, and vocabulary words, and wraps the data in a responsive HTML template.
- **Node 4: Gmail (Send a message)**
 - **Function:** The delivery mechanism.
 - **Configuration:** Authenticates via Google Cloud OAuth2 to send the generated HTML payload to a predefined list of recipients (parents/guardians).

3. How It Works (Execution Flow)

When demonstrating the workflow, you can walk through this step-by-step lifecycle of a single execution:

1. **Initiation:** The Schedule Trigger fires off the workflow at the scheduled time.
2. **Data Ingestion:** The workflow reads the master Excel sheet. By reading it fresh every day, any updates or changes made by teachers or parents to future dates are automatically captured without needing to update the n8n workflow.
3. **Intelligent Parsing (The JavaScript Engine):**
 - *Date Matching:* The script generates today's date and scans the "DAY / DATE" column of the spreadsheet to find the matching block of rows.
 - *Deep Column Scanning:* To account for messy data or shifted cells, the script forces a wide bounding box (scanning up to 30 columns deep) to ensure no assignments or remarks are accidentally cut off.
 - *Categorization:* It intelligently categorizes rows into Subjects (e.g., Mathematics, English), general Requirements, Maths Tables, and a day-specific Vocabulary Check.
4. **Template Generation:** The extracted raw text is injected into a custom HTML template. This transforms flat spreadsheet data into visual "cards" with color-coded borders, bold labels, and bulleted lists.
5. **Dispatch:** The Gmail node receives the HTML output and the success status, dispatching the email instantly.

Final Email received

